

Alex Pierron

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📍 Saint-Cloud, France

🔗 <https://alex-pierron.github.io/>

in Alex Pierron

🔗 <https://github.com/alex-pierron>

Education

PhD Institut Polytechnique de Paris [🔗](#), SAMOVAR Laboratory [🔗](#): AI researcher, PhD Candidate

Palaiseau, France
Oct. 2024 to Sept. 2027
3 years

- **PhD proposal:** Cybersecurity impact of AI Optimization in B5G networks.
- **Keywords:** Artificial Intelligence, Physical AI, Single & Multi-Agent Reinforcement Learning, Cybersecurity, Backdoors, 6G Networks, Physical Layer.
- **Abstract:** This PhD studies the security of AI-driven 6G wireless networks, with a particular focus on Reinforcement Learning and Reconfigurable Intelligent Surfaces, aiming to identify vulnerabilities and develop effective countermeasures.
- **Research Environment:** PhD within the [PEPR Future Networks](#) [🔗](#) program (France 2030 Label), combining rigorous academic research with interdisciplinary and industry-oriented collaborations.



MS Paris-Saclay University, Master Of Science, Mathematics and Artificial Intelligence [🔗](#)

Paris, France,
Sept. 2022 to Sept. 2024
2 years

- Master directed by the **University Mathematics Department** and shared with **CentraleSupélec**. 2nd year courses shared with the [MVA](#) [🔗](#) master's program at **ENS Paris-Saclay** and the StatsML master's program.
- **Most Relevant Coursework:** *Advanced Supervised & Unsupervised Methods, Graphical Models: Discrete Inference and Learning, Guidelines in Statistical Learning, Object Recognition and Computer Vision, Online Learning: link with Optimization and Games, Optimization, Sequential Learning, Statistics in Large Dimensions, Theory and Applications in Reinforcement Learning, Theoretical Foundations of Deep Learning.*



BS Paris-Saclay University, Double Bachelor of Science in Mathematics and fundamental Physics

Paris, France,
Sept. 2019 to Aug. 2022
3 years

- **Selective & intensive double bachelor's degree in mathematics and physics** directed by the university Mathematics and Physics Departments. Final year of physics shared with **ENS Paris-Saclay** and the fundamental physics "Magistère" program at Paris-Saclay University.
- Most Relevant Physics coursework: *Analytical Mechanics, Electromagnetism, Fluid Mechanics, Linear and Non-Linear Optics, Quantum Mechanics, Statistical Physics*
- Most Relevant mathematics coursework: *Algebra, Differential Calculus, Integration, Ordinary & Partial Differential Equations, Probability, Signal Processing.*



Publications

CONF Pierron, A., & Lahire, T. **A Survey of Reinforcement Learning for Autonomous Air Combat: Current Progresses and Limitations.** In *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems*. [DOI](#) [🔗](#)

AAMAS 2026
Paphos, Cyprus
May 25-29, 2026

CONF Pierron, A., Barbeau, M., De Cicco, L., Rubio-Hernan, J., & Garcia-Alfaro, J. **A Fairness-Aware Strategy for B5G Physical-Layer Security Leveraging Reconfigurable Intelligent Surfaces.** In *18th International Symposium on Foundations and Practice of Security*. **Best Paper Award.** [DOI](#) [🔗](#)

FPS 2025
Brest, France
November 25-27, 2025

Experience

Dassault Aviation : AI Researcher in Reinforcement Learning, Intern

Saint-Cloud, France
Apr. 2024 to Sept. 2024
6 months

- **Internship topic:** AI for collaborative air combat: Multi-Agent Reinforcement Learning (MARL).
- **Keywords:** Artificial Intelligence, Physical AI, Air Combat, Deep Learning, Multi-Agent Reinforcement Learning, Applied Research.
- **Main tasks:** Literature review on RL/MARL and open-source resources, development of realistic Python/C++ RL environments for AI-driven tactical scenarios.



CNRS, Signals and Systems Laboratory : AI Researcher in Computer Vision, Intern

Gif-Sur-Yvette, France
Mar. 2023 to Jul. 2023
5 months

- **Internship topic:** Classify small databases using the knowledge of larger ones by Few Shot Learning.
- **Keywords:** Artificial Intelligence, Deep Learning, Few Shot Learning, Meta-Learning, Optimization, Image Processing, Classification.
- **Main tasks:** Research and implement few-shot learning methods in PyTorch, evaluate them on geological texture datasets and analyse the results to identify future research directions.



CNRS, IJCLab : Research Assistant, Intern

Orsay, France
Oct. 2020 to Jan. 2021
3 months

- **Description:** Part-time internship at the ThomX particle accelerator, focusing on camera calibration and image processing for measurement optimization.

Core Competencies

- Optimization, Probability, Statistics, Machine Learning, Deep Learning, Physics.
- Physical AI, Reinforcement Learning, Meta-Learning, Online-Learning, Few-Shot Learning, AI for Science.
- Understanding, modelling & solving problems. Use & clearly explain theories and results from scientific articles.
- Develop numerical tools to validate the concepts with practical applications.
- Collaborate & deliver in high-stakes academic/industrial research environments.
- Interdisciplinarity, Scientific Curiosity, Rigour, Proactive.

Main Practical Skills

Languages **English: Fluent - C1** Cambridge English Certificate : L&R General English - C1+, ID: KZW58-VFJCS
French: native tongue

IT Skills:

- Advanced proficiency in **Python**, with expertise in libraries such as **NumPy**, **Scikit-Learn**, **PyTorch**, **JAX** & **Multiprocessing** to optimize computation performance during parallel computing.
- Proficient in **R**.
- Familiarity and practical experience with **C++** & **CUDA**.
- **Git** for organizing collaborative work and **Slurm** for running programs on supercomputers.
- **LaTeX** & **Typst** for writing scientific papers and technical documentation, **Inkscape** for scientific illustrations.

Software: Windows & Linux, Visual Studio, R Studio, Anaconda, Docker, Github, Zotero